PregAssist: Pregnancy Support Mobile Application for Pregnant Mothers and Doctors

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***Abstract*—Pregnant care involves constant follow-ups, timely risk detection, and successful interactions between mothers and medics. This paper introduces PregAssist, a mobile integrated support system that is a synthesis of four products, including fetal health decision support, physical health risk prediction, AI-based mental health monitoring, and AR-driven emergency training. The fetal health module provides explainable and offline-capable cardiotocography (CTG) classification to assist clinical decision-making. XGBoost classifier with SHAP-based interpretability is used to analyze physical risks to generate individual recommendations and alerts. Mental health assessment integrates questionnaire-based indicators with CNN-driven facial emotion recognition, while federated learning preserves data privacy through on-device training. AR-based deterministic expert system offers protocol-adherent emergency simulations to improve preparedness activity when facing high-level of risk. Through the convergence of predictive analytics, explainable AI, privacy preserving learning, and training design, PregAssist assists in proactive maternal attention in urban and resource constrained environments via combining a hybrid mobile architecture.**

Keywords— Fetal Health Classification, Decision Support System, Maternal Mental Health, Explainable AI, Federated Learning, Augmented Reality, Rule-Based Expert System.

I. Introduction

Care of pregnancy involves constant attention of the pregnancy, identification of risks, and effective communication. The proposed PregAssist mobile application will offer solutions to these needs by incorporating four fundamental elements into it, including AI-based mental health monitoring, fetal health classification decision support system, physical health monitoring, and an emergency training module based on AR and AI. Collectively, these elements offer full-scale care to the expectant mothers and physicians through the integration of smart analytics, real-time data, and interactive instructions. PregAssist is created with a view of helping in routine antenatal care as well as urgent circumstances making it accessible even in resource-stricken environments. Clinical decision support is one of such, helping to decrease variability in diagnoses, and fetal monitoring is one of the priorities where intelligent assistance enhances the clinical information perception.

Fetal health surveillance represents one of the fundamental components of antenatal and intrapartum care because early detection of fetal compromise is associated with fewer negative consequences. Cardiotocography (CTG) is the most widely used method for this, recording fetal heart rate patterns and uterine contractions to support clinical assessment [1]. Manual CTG interpretation is, however, subject to inter-observer variability as it depends too much on the experience of the clinician in borderline cases. In order to decrease subjectivity, computerized analysis and machine-learning-assisted testing have been considered, and they prove that models based on data enhance the clinical decision-making process provided that they are consistent with significant physiological parameters [2], [3]. It is based on this that this study introduces a mobile-integrated decision support system (DSS) classifying fetal health as normal, suspicious and pathological. It is offline and explainable, which allows clinicians to assist in a wide variety of healthcare environments.

Although clinical decision support enhances regular monitoring, overall care should also be provided to guarantee that there is readiness in the realization of life-threatening emergencies. To minimize the time lag between diagnosis and immediate response, PregAssist can be enhanced with an interactive emergency training module with guided and immersive simulations to enhance user preparedness.

Technologically, the system is based on a hybrid mobile architecture that combines a native logic layer to an Augmented Reality engine of high fidelity. The simulation engine knowingly uses a deterministic rule-based expert system instead of probabilistic machine learning because the critical aspect of clinical safety is the priority. This ensures that there are no protocol adherence errors whatsoever and completely remove the unpredictable black box errors. The structure operates on the concept of deep-linking to transfer users directly to offline-enabled situations.

Besides physical emergencies, psychological wellbeing has a profound impact on the outcome of pregnancy. Mental health disorders among mothers are very common and mostly unaddressed because of the stigma and access to specialized care [4], [5]. Even though conversational agents powered by AI may be scalably supportive [6], the current solutions are frequently based on one-modality processing and centralized interpretation, which concerns privacy and transparency. Recent research emphasizes the importance of privacy-preserving and interpretable emotion-aware frameworks, including federated learning [7], [8]. In order to fill these gaps, the current paper suggests a privacy-conscious and multimodal conversational assistant that combines an adaptive LLM with text and face emotion recognition. It delivers personalized, emotion-aware responses and employs federated learning for secure on-device updates.

In addition to mental health, constant check-up of physical health is essential to avoid complications. Traditionally, medical practitioners keep track of such problems as high blood pressure by observing patients at the hospital during appointments, which is not real-time. But with new advances in machine learning, predictive systems have become possible so that they can analyze clinical data continuously to detect risks earlier. The studies show that machine learning algorithms are capable of assessing maternal risk effectively using physiological measurements, including blood pressure, blood sugar, body temperature, and heart rate [9], [10]. In this study, a predictive module for pregnancy risk assessment was developed based on these vital maternal health variables. The approach began with training a Random Forest model to evaluate feature influence, while the final model employed Extreme Gradient Boosting (XGBoost) to maximize prediction accuracy and reliability [11].

II. Literature review

Recent advancements in artificial intelligence, mobile health (mHealth) technologies, and immersive learning systems have significantly transformed maternal healthcare. Current studies provide references to the increased usage of AI to predict mental health, machine learning to monitor fetuses and mothers, and augmented reality to train medical specialists. These papers prove how clever analytics can be included to boost the results of pregnancy. Nonetheless, the majority of solutions available consider the individual functionalities instead of offering a single platform between mothers and healthcare professionals. In this context, fetal surveillance is a primary clinical concern, which has led to a significant amount of effort aimed at enhancing the CTG analysis. A CTG automation has developed out of early rule-based computerized systems that unify signal features [3] to more recent machine-learning pipelines, which are powerful by learning on structured, preprocessed features [2].

Although these improvements have been made on accuracy, there are still adoption barriers. Most of the high-performing models are black-box models, which restrict the clinician in explaining the decisions and makes high-stakes use untrustworthy [12], [13]. The explainable AI literature focusing on healthcare has also highlighted the fact that clinical acceptance improves in case the explanation is in human readable forms and is consistent with domain concepts [14]. Moreover, cloud inference is commonly used in many CTG decision-support implementations, and this is not feasible in low-resource settings. Such shortcomings encourage the design of lightweight offline-capable fetus DSS systems with intrinsic interpretability.

In addition to supporting diagnosis, researchers also note that it is important to enhance user preparedness to high impact yet rare emergencies. Immersive technologies were found to be valuable in enhancing health literacy and acquisition of skills [15]. Nevertheless, the high-fidelity Augmented Reality is still restricted to professional clinical training [16]. The resultant gap is an urgent need of available, consumer-friendly tools that can enable laypersons to overcome obstetric crises in a safe manner. The storage capacity of devices on the application is often a barrier to the deployment of the applications in resource-constrained environments [17]. Moreover, using predictive machine learning as the means of acute self-triage creates unacceptable false-negative risks. As a result, predictable rule-based systems continue to be critical to ensuring that clinical practices are adhered to through to the letter [18], eliminating the risk of stochastic inaccuracies in the event of an emergency.

Maternal mental health needs constant help as well in comprehensive care of pregnancy. The disorders are still very common but often undiagnosed because of stigma and lack of access to care which adversely impact maternal and fetal wellbeing [4], [5]. Recent developments have brought in conversational agents to offer availability of digital assistance [6]. Nevertheless, the current systems tend to use one-modality data and centralized processing, which is subject to privacy and transparency risks. Studies have demonstrated that multimodal emotion recognition, which is a mixture of text and visual stimuli, enhances the reliability of emotional states [7] [8]. Furthermore, privacy-preserving frameworks such as federated learning allow emotion models to be trained locally on user devices, securely protecting sensitive personal data [7], [8].

Alongside psychological assistance, machine-learning-based predictions of risks are applied to monitor the physiological status of the mothers. Physiological measurements include blood pressure, blood sugar, body temperature, and heart rate, which are very important and can be utilized as predictive healthcare systems [9]. Different classification algorithms were used on maternal risk prediction [10]. Random Forest is an ensemble learning model that is commonly used to learn the complex associations among clinical characteristics [11]. In the recent past, boosting algorithms including XGBoost have come into focus as they are well-known because of their high predictive accuracy and efficiency in operating structured clinical information [19], [20].

III. METHODOLOGY

The proposed PregAssist system is an overall mobile-based pregnancy support platform, which incorporates four intelligent features that can be used to assist both pregnant mothers and healthcare professionals. The system includes: (A) a decision support for fetal health classification using cardiotocography (CTG) data, (B) an AR and AI-driven emergency training module for pregnancy-related critical situations, (C) an AI-based mental health monitoring and risk prediction module and (D) a physical health monitoring and personalized wellness guidance module, for early detection of risks. All these elements are united in a single mobile application framework that will allow safe data processing, real-time analytics, and a smooth exchange of information between mothers and doctors. Fig. 1 represents the general system architecture.

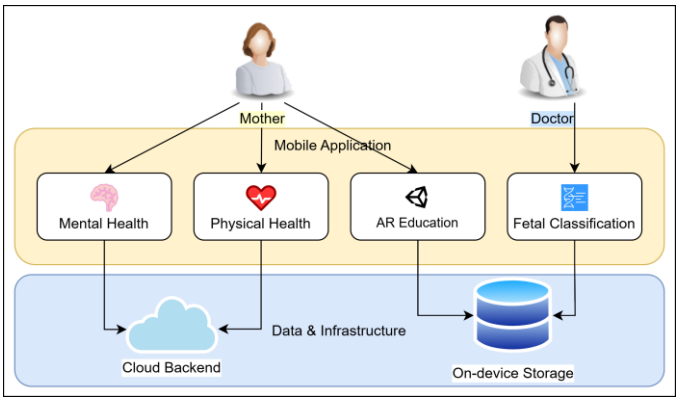

Fig. 1. System Diagram

A. *Decision support for fetal health classification using cardiotocography (CTG) data*

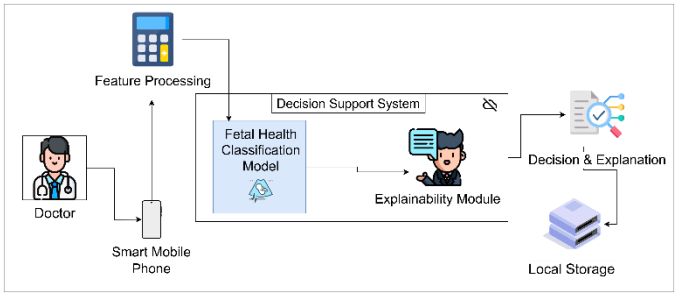

Fig. 2. Fetal Health Classification Diagram

Within the suggested DSS workflow, the obstetricians would choose the length of a CTG segment (e.g., 10-30 minutes) and input the measurements that can be regularly monitored on bedside CTG monitors or printed CTG report cards. There are inputs of the baseline fetal heart rate (FHR), the minimum and maximum FHR and the number of events occurred in the segmental choice: accelerations, uterine contractions, and mild, severe, and protracted decelerations. Since the source data model the various event variables as rates per second (small integer values), the application takes the naturally occurring (counts in a time period) and transforms it into rates per second by normalizing it with the chosen segment period. This does not imply a degradation of information but simply the preservation of the same clinical events but in a consistent numeric format that will be congruent with model expectations and current practice of CTG reading in the real world.

An open CTG fetal health tabular format dataset [21] was used to train the model and it contains 2126 records and 22 columns. Data cleaning eliminated duplicate samples and records that had missing data to eliminate noise. As some of the dataset attributes cannot be measured directly through visual interpretation, a strategy of features that can be conveniently handled manually was used by favoring clinically interpretable parameters and estimated some of the histogram summary statistics using baseline FHR to ensure that the feature would be compatible with inputs entered by doctors. Many algorithms were tried and the one that was chosen because of the most consistent overall performance with this feature set was the Random Forest. In case of offline mobile inference, the trained "joblib" model was translated into "ONNX" to allow the deployment on other frameworks without accuracy compromise. Explainability is generated in-device with a lightweight method: a distilled "shadow" surrogate creates justifications readable by clinicians, and counterfactuals provide explanations on the minimal changes in inputs that may cause the predicted classification, without incurring the huge SHAP-style runtime expenses.

B. *AR and AI-Driven Module for Personalized Pregnancy Emergency Training*

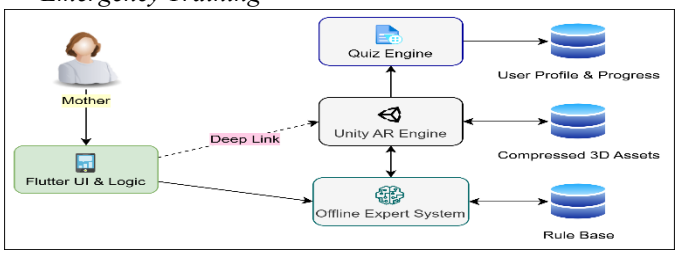

Fig. 3. System architecture of the AR and AI-Driven module

A Hybrid Mobile Architecture is used in the proposed system, which will operate a native Flutter environment with a core logic and a nested Unity 3D engine with visualization. This two-layer solution eliminates the computation overhead of a monolithic game engine, as it supplies the computational trade-off between fast data entry and high-quality rendering. By decoupling the lightweight triage interface from heavy Augmented Reality (AR) assets, the application maintains the low-latency responsiveness critical during obstetric emergencies.

To ensure a smooth transition between these two different environments, the architecture employs a special purpose Serialized Intent Passing protocol with the "FLAG_ACTIVITY_CLEAR_TASK" directive of Android. This procedure dictates the operating system to clean the prior Unity condition in advance of setting up particular AR scenes to prevent crashes on resources-limited machines. More importantly, this decouples the Dart based Intelligent layer and the C# based Visualization layer so that medical logic changes

can be made without having to reregularly reload huge bundles of 3D assets.

Clinical safety is given precedence; the inference engine substitutes probabilistic machine learning with a deterministic Rule-Based Expert System which has Forward Chaining. The system can achieve an explainable AI with zero stochastic error by rigidly coding WHO and ACOG guidelines into a decision tree. Lastly, diagnoses are based on the Symptomatic Feature Aggregation which confirms the quantitative vitals and the qualitative symptoms in real-time to provide a complete, error-free physiological snapshot.

C. AI based mental health monitoring module

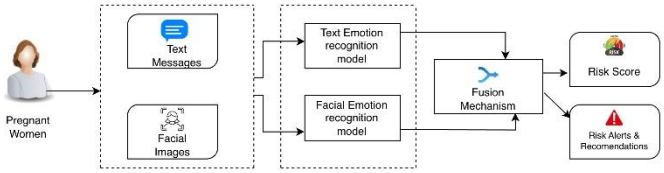

Fig. 4. System architecture of the AI based mental health module

The intended system will carry out a multimodal emotion conscious backend that aims to support the psychological wellbeing of expectant mothers using an AI-based chatbot. The system is developed as a Python Flask REST API and integrates three main sources of emotional indicators: facial emotion recognition, text-based emotion detection, and self reported questionnaires. For facial emotion recognition, the CNN model is trained using the AffectNet dataset, selected because CNN is highly effective in image classification tasks due to its ability to automatically extract spatial facial features and recognize complex emotional patterns. AffectNet images are preprocessed through face detection and cropping using Haar Cascade in OpenCV, followed by resizing to 48×48 pixels and normalization to a 0–1 range to ensure consistent input for model training. During interaction, a pregnant user communicates with the AI agent through chat, and the system continuously monitors emotional cues throughout the conversation. The uploaded facial image is analyzed and classified into anger, fear, happy, neutral, sad, surprise, and disgust categories.

At the same time, text messages provided by users are processed using TF-IDF [7], [8]. vectorization, and emotion is predicted using a Logistic Regression classifier trained on the GoEmotions dataset. Logistic Regression was selected because it performs efficiently on TF-IDF-based textual data and enables fast real-time classification suitable for chatbot interactions. Before training, GoEmotions text data is preprocessed by converting text to lowercase, removing punctuation and unnecessary symbols, and transforming cleaned text into numerical TF-IDF vectors. Gender classification is supported using the Human Images Dataset (Men and Women) from Kaggle, where images are resized, normalized, and labeled before use to maintain uniformity in prediction. The system then determines the final emotional state by combining facial and textual predictions and selecting the emotion with the highest confidence score. The AI chatbot dynamically adapts its responses based on the detected emotional condition and provides empathetic support when negative emotions are identified. Further, a self-reported questionnaire score is combined with facial emotion confidence through an equal-weight scoring algorithm to generate a final risk score. To ensure privacy preservation [5], the system incorporates federated learning, enabling local model updates on user devices while sharing only encrypted model parameters with the central server, preventing raw sensitive data from leaving the device.

D. Physical health monitoring and personalized wellness guidance module

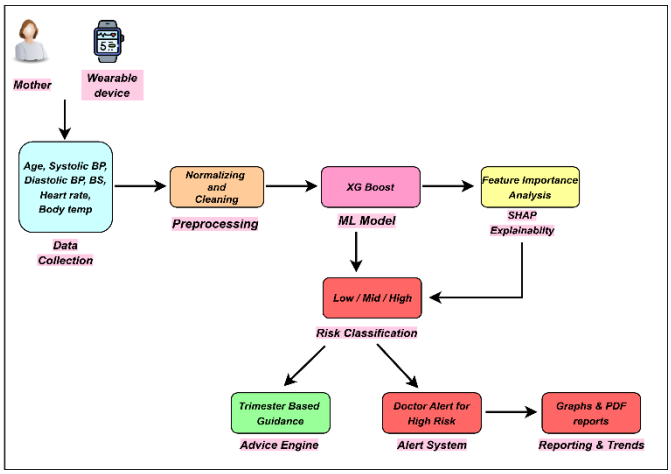

Fig. 5. Component Architecture of the Pregnancy Risk Prediction Module

The proposed pregnancy risk prediction system utilizes machine learning techniques to classify maternal health risk levels based on physiological parameters. The dataset used in this study was obtained from a publicly available Kaggle maternal health dataset [22], consisting of 1,015 records with physiological parameters such as age, systolic blood pressure, diastolic blood pressure, blood sugar level, body temperature, and heart rate. These parameters are commonly used indicators in maternal health monitoring and are strongly associated with pregnancy complications such as hypertension and gestational diabetes. RiskLevel is the target variable of the dataset and it has three classes that include Low Risk, Mid Risk, and High Risk.

Data preprocessing to prepare the data to be analyzed by the machine learning models was carried out before training them. This was done by dealing with missing data, normalizing numerical data, and transforming labels into numbers that can be classified by classification models. Exploratory data analysis was conducted as well to learn the connections between the input features and the levels of risk of pregnancy.

In the first part, a Random Forest classifier was trained to be used as a baseline prediction model and to determine the importance of features in the data set. Random Forest will create several decision trees and combine their outputs to enhance the precision and minimize overfitting [11]. After evaluating the baseline results, XGBoost was selected as the final model due to its ability to improve prediction accuracy and effectively capture complex relationships in structured healthcare data [19]. Existing maternal health applications such as BabyCenter and Pregnancy+ mainly focus on basic tracking and monitoring,

whereas the proposed component provides advanced features including explainable AI, continuous monitoring, and early warning alerts.

IV. Results and discussion

This section analyzes the performance, diagnostic reliability and systems efficiency of the integrated PregAssist modules.

During the development of the decision support system for fetal health classification, multiple machine learning classifiers were evaluated using a clinically aligned cardiotocography (CTG) feature set. Linear models like the Logistic Regression had lower generalization and non-linear approaches enhanced test accuracy. Random Forest as indicated in Table I has the best overall balance between learning capacity and generalization. Even though the original default Random Forest setup exhibited evident overfitting (99% training accuracy vs. 88% test accuracy), such regularization measures as restriction of tree depth, minimum sample required to use splits, and class balancing were implemented. This minimized model variation and smaller train-test gap (92% train accuracy) without compromising on an 88% test accuracy, leading to a stable model that can be used in the real clinical setting.

TABLE I. Performance Comparison of CTG Classification Models

Model	Train Accuracy	Test Accuracy
Logistic Regression	0.81	0.78
SVM (RBF)	0.83	0.81
KNN (K=5)	0.91	0.86
Decision Tree	0.89	0.82
Random Forest	0.92	0.88

Treating acute care of the mother, the test of the AR and AI-interactive emergency training module showed that a deterministic rule-based system can provide better obstetric triage safety than probabilistic machine learning. The inference engine tested on a synthetic validation set (N=200) had a 100% sensitivity value, and a false-negative value of 0.0, which means that no life-threatening emergencies were omitted. Although there was a deliberate false-positive of 5.5 percent, this conservative bias is acceptable within the high-stakes medical settings as compared to the 8-12 percent error rates found with the stochastic models [23]. The hybrid mobile architecture was also confirmed by performance profiling. Decoupling the native logic layer and Unity engine allowed the latency of deep-linking to decrease to 1.42 seconds (far lower than the desired 2.0s), and the maximum memory consumption leveled off at 240 MB, with an effective 32 FPS. This is very fast and storage efficient compared to monolithic game-engine applications [16]. Regarding the knowledge that was gained pedagogically, the percentage of knowledge gain in emergency procedure identification was 42% and the System Usability Scale (SUS) of 86/100 validated the reduced cognitive load. This establishes that the immersive spatial interaction is highly effective to construct situational awareness compared to the non-immersive text-based app [24].

To test the continuous psychological support, the AI-based mental health monitoring module was evaluated by comparing single-modality detections with the proposed multimodal fusion strategy. The system combines facial emotion recognition using CNN with TF-IDF [6]. and text emotion classification using Logistic Regression [4]. As shown in Table II, although individual modalities provide useful predictions, they are limited by environmental conditions such as lighting and input clarity. Compared with existing systems such as questionnaire-based maternal mental health applications and rule-based chatbot systems like Moment for Parents, which mainly rely on self-reported text interactions, the proposed multimodal fusion mechanism improves detection reliability by integrating both facial and text emotions, achieving 91% accuracy with a reduced false negative rate of 8%. This is critical in mental health monitoring, where failure to detect negative emotional states may delay timely support, while the adaptive chatbot enables more personalized interventions.

TABLE II. Performance of Emotion Detection Approaches

Metric	Facial Emotion (CNN)	Text Emotion (TF-IDF + LR)	Proposed Multimodal Fusion
Sensitivity	86%	84%	92%
Specificity	83%	81%	90%
Accuracy	85%	83%	91%
False Negative Rate	14%	16%	8%

Finally, the physical health monitoring module evaluated the performance of Random Forest and Extreme Gradient Boosting (XGBoost) for pregnancy risk prediction. The accuracy and F1-score of the Random Forest model were approximately 81.3, which is sufficiently close to accurately detecting the level of the risk depending on physiological indicators (blood pressure, blood sugar, body temperature and heart rate). This is consistent with the past research findings where the ensemble learning models have been found to be reliable in healthcare prediction [11], [25]. Nonetheless, as Table III describes, XGBoost was more effective in all the metrics of classification. With the advantage of a boosting mechanism, which progressively enhances weak learners, XGBoost achieved an accuracy of 85.22% accuracy. These findings confirm that advanced boosting-based algorithms offer superior predictive capabilities for maternal health risk assessment compared to traditional classification models, consistent with recent healthcare literature [9], [10], [19], [20].

TABLE III. Performance Comparison of Risk Prediction Models

Model	Accuracy (%)	Precision (%)	Recall (%)
Random Forest	81.28	81.85	81.28
XGBoost	85.22	85.12	85.22

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